Milestone 1

Problem Definition & Literature Review [October 3]

1. Define the objectives of the project:

Objective: To build a Clothing Assistant.

A chatbot designed to answer questions from store policy, previous purchases of customers, and suggest outfits based on query given and customer preferences.

Part A: Store policy + Customer help desk

The chatbot will answer questions from store policies like what is the return policy, upcoming discount info, and so on. Additionally the customer can select a previously bought item and ask something like whether I can return or replace the product. Now the agent will check the return or replacement policy of that specific product and reply accordingly.

Part B: AI stylist

- The agent will be fed context about the customer preferences (liked outfits, size, etc). Without any query given the agent should concentrate on these preferences and suggest clothes. An additional advanced option could be suggesting clothes based on the skin tone of the customer if shared.
- Initially to get customer preferences and serve well a quiz will be designed.
- Given a query the agent should consider both customer preferences and query given and suggest clothes. (An example query would be 'suggest me an outfit for a graduation party')
- To prevent hallucinations the agent will only suggest products from the company catalogue. Also the agent should stick to the job of answering from company policy or suggesting outfits.
- The agent should have knowledge about fashion like which color shirt matches with which color pants, on which occasions what outfit is appropriate, etc,...

2. Review of existing solutions, baselines, and benchmarks:

a. **Existing solutions -**

   - **Zalando Assistant** (Zalando launched a chat/assistant experience powered by LLMs and in-house models to help product discovery). Their public reports focus on business KPIs (clicks, wishlist adds, conversion uplift), not BLEU/F1.
     Example: Zalando reported product-click and wishlist uplift when deploying their assistant.
   - **H&M "Virtual Stylist"**, experimental assistants / chatbots for style recommendations, virtual try-on etc.
   - **Klarna's "Shop-with-me / AI Style advisor"**, Style recommendations, curated product browsing, conversational interface in app / web.
   - **Amazon's Rufus**, a generative AI-powered conversational shopping assistant built by Amazon. It's trained on *Amazon's product catalogue*, *customer reviews*, *community Q&A*, *listings info*, plus "relevant info from across the web."

b. **Current baselines and benchmarks -**

   - **Fashion-IQ (conversational / relative feedback image retrieval)** — a public dataset created to study dialog-style fashion image retrieval. **Recall** (R@10/R@50 based metrics) ranging from ~**39%** to **~66%.**

     We could use **Avg Recall@10 / Avg Recall@50** as the baseline metric for retrieval tasks.

     [The **Recall@10** is the number of relevant items in the top 10 results divided by the total number of relevant items in the entire dataset for a given query.]

   - **SimpleMTOD**: **BLEU ≈ 0.327**

     [BLEU (Bilingual Evaluation Understudy) is an algorithm for evaluating the quality of text which has been machine-translated from one natural language to another.]

3. Gaps and Opportunities:

- Based on existing solutions we found, **the agents did not suggest clothes from a catalogue** rather it had multiple data sources (multiple store websites) which made the agent third party.
- **The agents did not stick to the customer request** (when asked for a pant, the agents suggested other products too)
- **The agents did not have much fashion sense**. Not always relevant products were shown. For example, when asked to suggest a pant for an already available red shirt, the agent did not pair a proper pant. Actually the pants suggested didn't even take into account the red shirt.
- **We didn't find any agents which requested for size** and filtered based on that too. The agents suggested a clothe and after the customer selects the product it showed sizes available.
- **The chat history is not maintained properly**. For example, we requested an outfit for a business meeting, and the agent suggested some blazers. Next we gave a prompt 'need a blue one'. Here the bot forgot about the occasion 'business meeting' and suggested even T-shirts.